

Security risk assessment report

Part 1: Select up to three hardening tools and methods to implement

1. Multifactor authentication.
2. Password policies.
3. Firewall maintenance.

Part 2: Explain your recommendations

1. Multifactor authentication (MFA)
 - Implementing MFA requires users to verify their identity using two or more authentication factors. This provides a vital layer of defense, even if internal credentials are leaked, shared or guessed via automated brute force, an unauthorized actor cannot access internal systems without the secondary physical verification factor.
2. Password policies
 - The organization must deploy strict password policies aligned with current NIST guidelines to eliminate the use of shared accounts and default administrative passwords. Policies must mandate that database administrative credentials be modified immediately upon deployment to eliminate automated guessing vectors. Furthermore, individual employee accounts must be enforced to ensure non-reputation, alongside system level salting and hashing mechanisms to secure stored authentication data.
3. Firewall maintenance
 - The security team needs to perform periodic firewall maintenance to establish and update comprehensive rules for filtering inbound and outbound traffic. Currently, the lack of traffic inspection mechanisms allows for uncontrolled network access. Configuring stateful packet filtering and blocking unnecessary ports will effectively isolate internal database assets from the external Internet, preventing direct access attempts from suspicious IP addresses and limiting unauthorized data leakage channels.

